

Koh Seungbeen

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RESEARCH INTERESTS

Protein structural biochemistry; GPCR and membrane receptor biochemistry; receptor–ligand recognition; NMR- and cryo-EM-assisted structural analysis; structure-guided drug discovery; AI-enabled protein design.

EDUCATION

Yonsei University, College of Life Science and Biotechnology

Seoul, Republic of Korea

B.S. in Biochemistry

GPA: 3.91 / 4.30 | Expected Graduation: August 2026

Gwangju Science Academy for the Gifted

Gwangju, Republic of Korea

Graduated from the 34th class

- Presented an R&E poster, “Antibacterial Effects of Mushroom Fermentation Extracts Against Pepper Pathogens,” at the 8th Joint Presentation Meeting for Outstanding R&E Projects of Science Gifted Schools.

PUBLICATION

Koh, S., Park, M., Jung, D. J., Park, J.-H., & Lee, W. (2026). Expression, purification, and preliminary cryo-EM analysis of the human urotensin 2 receptor/urotensin 2 complex. *Protein Expression and Purification*, 242, 106961.

<https://doi.org/10.1016/j.pep.2026.106961>

RESEARCH EXPERIENCE

Structural Biochemistry & Biophysics Laboratory, Yonsei University

Dec 2023 - Jun 2026

Undergraduate Researcher / Manuscript Contributor | PI: Prof. Weontae Lee

Active bench research: Dec 2023 - Dec 2024; data analysis and manuscript preparation: Dec 2024 - Jun 2026

Project 1: Human urotensin 2 receptor (UR2R)/urotensin 2 (hU2) complex expression, purification, and preliminary cryo-EM analysis

- Led the experimental workflow for establishing recombinant expression, purification, and complex preparation of the human urotensin 2 receptor/urotensin 2 complex, a class A GPCR system relevant to cardiovascular and metabolic disease biology.
- Constructed human UR2R expression plasmids using the pFastBac/Bac-to-Bac system and performed baculovirus amplification, Sf9 insect cell expression, membrane preparation, detergent solubilization, TALON affinity purification, reverse Ni-NTA chromatography, HRV 3C cleavage, and ÄKTA-based SEC.
- Assembled the UR2R¹¹⁻³³⁶–hU2–mGqi–scFv16 signaling complex for cryo-EM sample preparation and contributed to preliminary cryo-EM characterization, including grid preparation, data collection/processing, and 2D class analysis as part of a collaborative workflow.
- Organized experimental data, prepared figures, and contributed to manuscript writing for the published study.

Project 2: Somatostatin analog structural analysis and receptor selectivity

- Led a structural analysis project investigating somatostatin receptor selectivity by integrating NMR data processing, peptide structure calculation, receptor–ligand interaction analysis, sequence alignment, and HADDOCK-based docking.
- Participated in NMR sample measurements and analyzed HSQC, NOESY, and TOCSY spectra of designed somatostatin analogs using POKY.
- Determined structural features of somatostatin analogs, compared their conformations with SST-14, and interpreted SSTR2 docking models to investigate receptor-recognition motifs associated with SSTR selectivity.

- Generated PyMOL-based structural figures and prepared the poster content, visual layout, and presentation for ICMRBS 2024.

Project 3: Human somatostatin receptor 5 (SSTR5) purification and structural characterization support

- Contributed to SSTR5 purification workflows in a draft-stage GPCR structural biology project, including membrane protein preparation, affinity purification, detergent exchange, reverse Ni-NTA chromatography, and SEC.
- Supported organization and interpretation of purification and complex-formation data for SSTR5–Gαi–scFv16 complexes.

Epigenogen, Yonsei University-affiliated AI Drug Discovery Company

Jun 2024 - Aug 2024

Research Intern

- Assisted with purification of ASH1, an epigenetic target implicated in acute leukemia, and evaluated ASH1-targeting compound treatment by western blot analysis.
- Gained exposure to biochemical validation workflows in an AI-assisted epigenetic drug discovery setting.

Epigenetics and Molecular Pharmacology Laboratory, Seoul National University

Dec 2024 - Feb 2025

Undergraduate Research Intern | PI: Prof. Chul-Hwan Lee

Project 1: H3K27M-mediated PRC2 inhibition in DIPG

- Reconstituted WT, H3K27A, and H3K27M histone octamers and purified H3K27A/H3K27M octamers using ÄKTA-based size-exclusion chromatography.
- Validated histone octamer fractions by SDS-PAGE and Bradford assay.
- Assisted with PRC2–dinucleosome interaction experiments and HMT assay workflows to evaluate H3K27M-dependent changes in PRC2 methyltransferase activity.

Project 2: Sex-specific epigenetic modifications in mouse microglia

- Performed CUT&Tag experiments using male and female mouse cortex and hippocampus samples to profile H3K4me3- and H3K27me3-associated chromatin states.
- Conducted qPCR-based quality control of CUT&Tag libraries using HoxA9, Oct4, and indexed primer sets.
- Gained exposure to epigenomic profiling workflows for studying sex-specific chromatin regulation in neurodegenerative disease-related contexts.

CONFERENCE AND POSTER PRESENTATIONS

Koh, S., Lee, D., Lee, D., Yun, J.-H., Song, J.-J., & Lee, W. (2024). Structural Analysis and Functional Dynamics of Somatostatin for Receptor Selectivity via NMR and Cryo-EM. Poster presented at the International Conference on Magnetic Resonance in Biological Systems (ICMRBS 2024), Seoul, Republic of Korea, August 18–23, 2024.

Koh, S. (2024). GPCR purification, somatostatin NMR structural analysis, and receptor selectivity. Three individual posters presented at Yonsei Biochemistry Fair, Yonsei University, Seoul, Republic of Korea.

NanoSEA Academic Club, Yonsei University, Seoul, Republic of Korea, 2024. Contributed to and presented two team-based literature review posters on CAR-T cell engineering and SynNotch-CAR T cell platforms.

TECHNICAL SKILLS

Protein Biochemistry and Membrane Protein Preparation: molecular cloning; pFastBac/Bac-to-Bac baculovirus expression system; baculovirus amplification; Sf9 insect cell expression; membrane preparation; detergent solubilization; GPCR purification; ligand-bound receptor and receptor–G protein complex assembly.

Protein Purification and Biochemical Assays: TALON metal affinity purification; Ni-NTA and reverse Ni-NTA chromatography; HRV 3C protease cleavage; ÄKTA-based size-exclusion chromatography; SDS-PAGE; anti-FLAG immunoblotting; Bradford assay; western blotting; compound treatment assays.

NMR, Cryo-EM, and Structural Analysis: POKY-based HSQC, NOESY, and TOCSY analysis; TopSpin-based basic 1D NMR analysis; NMR-based peptide structural analysis; cryo-EM grid preparation and preliminary data analysis exposure; PyMOL-based structural visualization; introductory HADDOCK docking; CLUSTALW/sequence alignment.

Chromatin Biology and Epigenomics: histone octamer reconstitution; PRC2–dinucleosome interaction workflow; HMT assay; CUT&Tag; qPCR-based library quality control.

Programming and Computational Skills: basic Python programming; coursework-based training in software programming and artificial intelligence algorithms.

HONORS AND AWARDS

Academic Excellence Recognition, Yonsei University Fall 2023; Spring 2024

Selected Named, University, and Foundation Scholarships 2020 - 2025

- Named/Foundation Scholarships: DB Kim Jun-Ki Cultural Foundation Scholarship; Kim Soon-Jeon Scholarship; Goseon Scholarship; Jeong Myung-Ok Scholarship.
- University Scholarships: Yonsei Scholarship; University-Designated Scholarship.

ACADEMIC ACTIVITIES AND SERVICE

NanoSEA Academic Club, Yonsei University Apr 2024 - Jun 2024; Sep 2024 - Nov 2024

Member

- Participated in student-led academic discussions and literature review activities in biotechnology, nanomedicine, immunotherapy, and synthetic biology.
- Presented two review posters on CAR-T cell engineering and SynNotch-CAR T cell platforms.

Korean Magnetic Resonance Society Meeting Jan 29-31, 2024

Participant | Gwangju, Gyeonggi-do, Republic of Korea

Make-A-Wish Korea Feb 2024 - Jun 2026

Volunteer

- Participated in long-term volunteer activities supporting children with critical illnesses and their families.

Community Service 2024 - 2026

Volunteer

- Completed approximately 162 total hours of volunteer service during undergraduate studies.

SELECTED COURSEWORK

Biochemistry; Biophysical Chemistry; Molecular Biology; Cellular Immunology; Epigenomics; Biostatistics and Bioinformatics; Artificial Intelligence Algorithms; Software Programming.